COMPUTER STUDIES NOTES

INFORMATION TECHNOLOGY

- **Information Technology** (IT**) refers to a set of tools for working with information in all aspects of managing and processing information**.
- This includes **keyboards, mice, screens, printers, modems**,
- **The purpose of a computer system is to convert data into information used for decision making.**

INFORMATION AND COMMUNICATION TECHNOLOGY (ICT).

- Communication technology consists of both physical devices and software that link various computer hardware components and transfer of data from one physical location to another.

COMPUTER SYSTEM

- **The term ''computer system''** refers to the whole of computer hardware components, peripherals and data communication equipment.
- **A computer** is an electronic device which functions under the control of a stored computer program.
- **A computer** is an electronic device which is capable of receiving information.
- **A computer** is an electronic device which accepts data as inputs stores it and processes it and gives out output called **information.**

HARDWARE.

- Computer hardware consists of the components that can be physically handled.
- Computer hardware refers to the physical parts or components of a computer that you can physically touch.
- These include the Central Processing Unit, the keyboard, Mouse, Printer, Scanner, Uninterruptible Power Supply, Monitors also known as the **Visual Display Unit.**
- The term ***device*** refers to any piece of hardware used by a computer, such as keyboard, monitor, modem, and mouse.
  PARTS OF THE COMPUTER SYSTEM
- Keyboard, mouse, central processing unit, visual display unit (VDU)

  - ***KEYBOARD***

  A keyboard is the mostly widely used device for inputting data into computers. It can be used to enter any kind of alphabetical or numerical data,

MOUSE

This is a hand- held device you use to move a graphical pointer on the screen. The buttons are used to select an icon (pictures representing options).

SYSTEM UNIT

- The system unit is the central part of a computer system.
- The most important of these components is the **central processing unit** or microprocessor, which acts as the **brain** of the computer.
- Another component is **random access memory** (RAM), which temporarily
- Hardware that is not part of the system unit is sometimes called a **peripheral device** such as a keyboard or a mouse.

THE CENTRAL PROCESSING UNIT (CPU)

- This is a device that is housed in the system unit box.
- The CPU contains the **Control Unit** (CU) and the **Arithmetic Logic Unit** (ALU) and **main memory.**
- The Control Unit, controls the movements of data.
- The ALU performs mathematical functions and logical decisions.
- An important feature of the CPU is the speed of its internal clock that is measured in **megahertz.**

SCANNERS.

- These are devices which can read printed or magnetic data and convert it into electronic signals to be sent to an attached computer.
- A scanner is an input device that is used for entering graphics and images such as a photograph.

PRINTER

- A printer transfers data from a computer onto paper.
- **A Printer is a device that produces text and graphics on a physical medium such as paper or transparency film.**
- These enable permanent records of selected information to be produced.
- Such paper copies of information are also referred to as **hard copy.**
- The two main types of printers are ***inkjet printers and laser printers.***
- Inkjet printers are the most popular printers for the home. They can print in **black** and **white** or in **full colour** and can produce high quality.

UNINTERRUPTIBLE POWER SUPPLY (UPS)

A UPS is used to safeguard your data against any data loss due to power cuts.

So what it means is that when power is cut your computer has a UPS installed, you can continue using your computer, though up to a certain limited extent. So this will allow you to save your documents in time.

MONITOR

- A monitor displays information in visual form, using text and graphics.
- The portion of the monitor that displays the information is called the **Screen.**
- Like a television screen, a computer screen can show still or moving pictures.
- **THERE ARE TWO BASIC TYPES OF MONITORS:**
- **CRT: (Cathode ray tube) monitors** and **LCD (Liquid Crystal Display) monitors.**
- Both types produce sharp images, but **LCD monitors have the advantage of being much thinner and lighter.**
- **CRT monitors** however, are generally more affordable.

SOFTWARE

- Software is the term that is used to describe a set of programmed instructions that tells the computer what to do.
- **Software is a set of instructions that tells the computer and its devices what to do.**
- **Software is the program that run the computer.**
- A logical set of instructions given to the computer is known as **a program.**
- This helps to interact with computer to perform a particular task.
- **Software is a set of electronic instructions that tells the computer how to do certain tasks.**
- **A set of instructions is often called a program.**
- **When a computer is using a particular program, it is said to be running or executing the program.**

TYPES OF SOFTWARE

There are two main types of computer software namely:

1. System Software
2. Application Software.

SYSTEM SOFTWARE

- System software is a set of instructions needed for a computer to work.
- This is a computer software that operate and control the computer hardware and application software.
- This is the software that runs the computer. It is commonly known as the **operating system.**

OPERATING SYSTEM

- This is software that manages or control computer hardware and software resources.
- An operating system is a collection of software that manages computer hardware resources and provides common services for computer programs.
- **An Operating system** is responsible for loading other programs into computer memory for execution although it is also a program that needs to be loaded into memory in order to run.

EXAMPLES OF OPERATING SYSTEM

- DOS (Disk Operating System),
- MS- DOS
- IBM
- Linux
- Unix
- Mac OSX
- Windows 1.0, windows2.0, windows 3.0, windows 3.1
- Windows NT, windows 95, windows 98, windows 2000, windows ME
- Windows XP, windows Vista, windows 7, windows 8.
- And the most recent operating systems which many are using now. Are windows 7 and windows 8?
- The operating systems used for running smartphones are IOS, Android, windows etc.

APPLICATION SOFTWARE

- This is a computer program that allows users to do everyday tasks.
- These are programs designed to perform specific tasks for users.
- **Application software tells the computer how to accomplish tasks the user requires, such as creating a document or editing a graphic image**

EXAMPLES OF APPLICATION PROGRAMS

- **WORD PROCESSING** SOFTWARE: allows users to create, edit and print text documents that contain text and graphics. Example, Microsoft word.
- **SPREADSHEET SOFTWARE**: with spreadsheet software, data is organised in rows and columns, which collectively are called a worksheet. Examples of spreadsheets: Ms Excel, Open Office
- **DATABASE SOFTWARE:** This software allows you to create and manage a database. A database is a collection of data organised to allow access, retrieval, and use of that data. Examples of database: structural Query Language), Ms Access, Oracle, Sybase, 4th Dimension, Paradox, Open Office base.
- **PRESENTATION SOFTWARE:** is used to create presentations that communicate ideas, messages and other information to a group through a slide show. You can use a clip gallery to enhance your presentation with clip art images, pictures, video clips, and audio clips.
- Examples of presentation software: Ms PowerPoint, Open Office.
- **DESK TOP PUBLISHING**: software is used to design and produce sophisticated documents such as newspapers, calendars, magazines.
- It is used to make magazines, posters, or an advertisement.
- Examples of DTP: Ms Publisher, Adobe PageMaker, Adobe InDesign, QuarkXPress.
- ***PAINT SOFTWARE***: is used to draw graphical images with various on screen tools. Examples: Paint brush, Tux paint.

COMPUTER- AIDED DESIGN (CAD): software assists in creating engineering,

CLASSIFICATION OF COMPUTERS

SUPER COMPUTER

- The most powerful and most expensive type of computer available- were developed to fill this need.
- Some applications required extraordinary speed, accuracy, and processing capabilities- for example, sending astronauts into space, controlling missile guidance systems and satellites, forecasting the weather, weapon design
- Exploring for oil, and assisting with some kinds of scientific research.

MAINFRAME

- Is the standard choice for large organisations such as hospitals, universities, large businesses, banks, and government offices- that need to manage large amounts of centralised data?
- Large general purpose computers with extensive processing, storage and input/output capabilities.

- They provide high- capacity data storage for Internet servers and because of this they are usually referred to as servers rather than mainframes.

MINICOMPUTER

- Minicomputers are multi-user systems that can handle the computing needs.
- The term minicomputer was introduced in the mid-1960s referring to relatively small and inexpensive computer systems.
- Typically used in a single department of an organisation and often dedicated to one task or shared by a small group.

MICROCOMPUTER

- Is a small computer system designed to be used by one person at a time
- PCs are small enough to fit on a desktop, inside a briefcase, or even inside a shirt.
- Microcomputer is a name given to a small low priced computer system as commonly found in homes and offices today.

TYPES OF COMPUTERS

THE DESK-TOP

- As the name tells, is usually located on top of the desk, with its tower (where its main memory is kept) placed either at the side of the desk of on the floor beneath the desk.

THE LAPTOP.
Also known as a notebook computer is small enough and light enough to sit on one's lap while working.
THE PALMTOP
Is the smallest of the three, and by far the lightest? It can easily fit into a jacket or a small bag.

What is data processing?

This is the conversion of raw data into useful information. Processing includes the following: calculating, drawing, searching, etc.

Data is **unprocessed information.**
Data: refers to raw facts, figures, text, etc., that is not processed and has no meaning.

Data can also mean unprocessed facts, figures, text, etc. that carry no meaning. Information on the other hand is processed data.

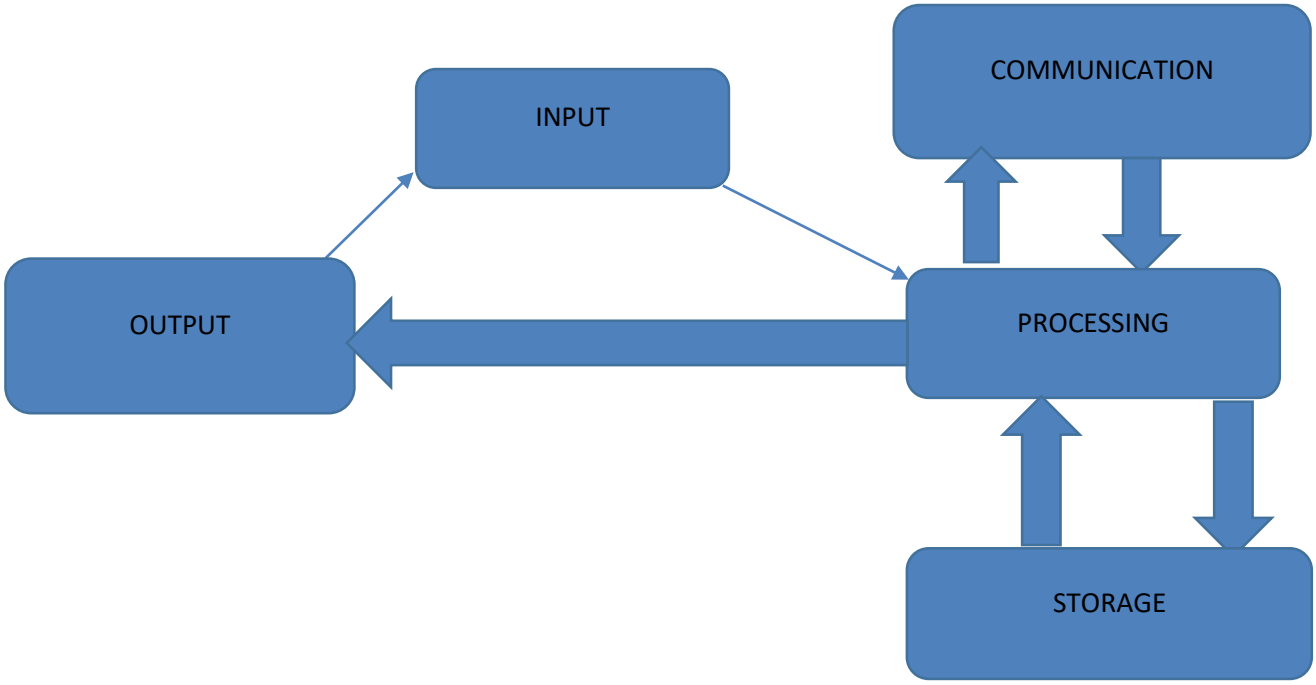
In other words, **data only becomes information when it has been converted or processed into meaningful information.**

Information is processed data in such a way that it is meaning and useful to the receiver.

PARTS OF A COMPUTER

MAIN PARTS OF A COMPUTER

- These are input, processing, output, storage and communication devices.
- A basic computer consists of the following hardware components: input, processing, output, storage and communication components.
- Input, output storage and communication devices that can be connected to a computer are also called **peripherals.**
- Physical components that are connected to the computer besides the CPU and Memory are referred to as **Peripherals**



INPUT DEVICES

- Input devices are used to enter data into the computer system.
- **An input device is used to enter data into a computer**.
- Examples of input devices are a **keyboard, mouse, digital camera, scanner and optical reader, touchpad, touch screen, trackball, joystick, stylus and digital pen, light pen, video camera, webcam and microphone.**
- The **keyboard** and **mouse** are the most commonly used input devices.

INPUT DEVICES

An input device is a device that takes in data into a computer.
It can also be a peripheral or piece of computer hardware equipment used to provide data and control signals to an information processing system such as a computer

A TRACKBALL
A pointing device that is an alternative to a mouse.
A trackball consists of a ball that is rotated to move the pointer on a computer screen. You can use a track ball when you have limited desk space. A device similar to a video camera.
It allows you to capture and send live pictures to other users.
MICROPHONE:
A device used to talk to people in different parts of the world
You can record your speech and let the computer convert it into text. It is a device to input sound information and can be used with a voice recognition system processes the information. It translates the speed into electronic signal. A

microphone allows a user to speak to the computer to enter data and instructions into the computer.

DIGITAL CAMERA. This allows you to take pictures and to store digital photographic images that can read by a computer. The images are transferred directly from your camera on to your computer.

LIGHT PEN: a pen- like device, attached to the computer by a cable, is placed on the screen and the computer determines its position from time when the illuminated data that builds up the whole VDU pictures passes under beneath the pen. The pen enables the user to point, write or draw directly on a sensitive screen or other surface using something similar to a pen.

TOUCH SENSITIVE SCREEN: this is a special type of VDU, which has a grid of light beams or fine wires criss- crossing the screen. When the screen is touched, the computer senses where you have pressed. The information is stored on the computer and touch screen is simply an interface. Touch screens are both input as well as output devices.

JOYSTICK: this is a device that enables that user to control movements on the screen by manoeuring a small lever. The lever can be moved in any direction from its zero position. It can also be made to produce faster movements on the screen by pushing it further from the zero position. An example of uses of joystick includes controlling objects in computer games to move objects on the screen and producing graphics.

VOICE RECOGNITION SYSTEMS: this is an input that converts the human voice into digital pattern, compares that pattern to a set of prerecorded patterns, and executes a command if a match is found. If no match is found, the system may ask the user for more input. It is also defined as a voice- recognition system may an input device that analysis and interprets the human voices to follow instructions.

MICR (MAGNETIC INK CHARACTER READER) magnetic ink characters are the strange looking numbers that appear at the bottom of cheques. Banks use MICR to read the numbers from the bottom of the cheques to obtain data such as account numbers and bank sort codes.

PC CAMERA (WEB CAMERA): A PC camera is a digital video camera attached to a computer. A PC camera allows home users to create a movie and take digital still photographs on their computer. With a PC camera, you can also have a video telephone call- where someone can see you while communicating with you.

(b). Optical Mark Readers and Optical mark Recognition (OMR)

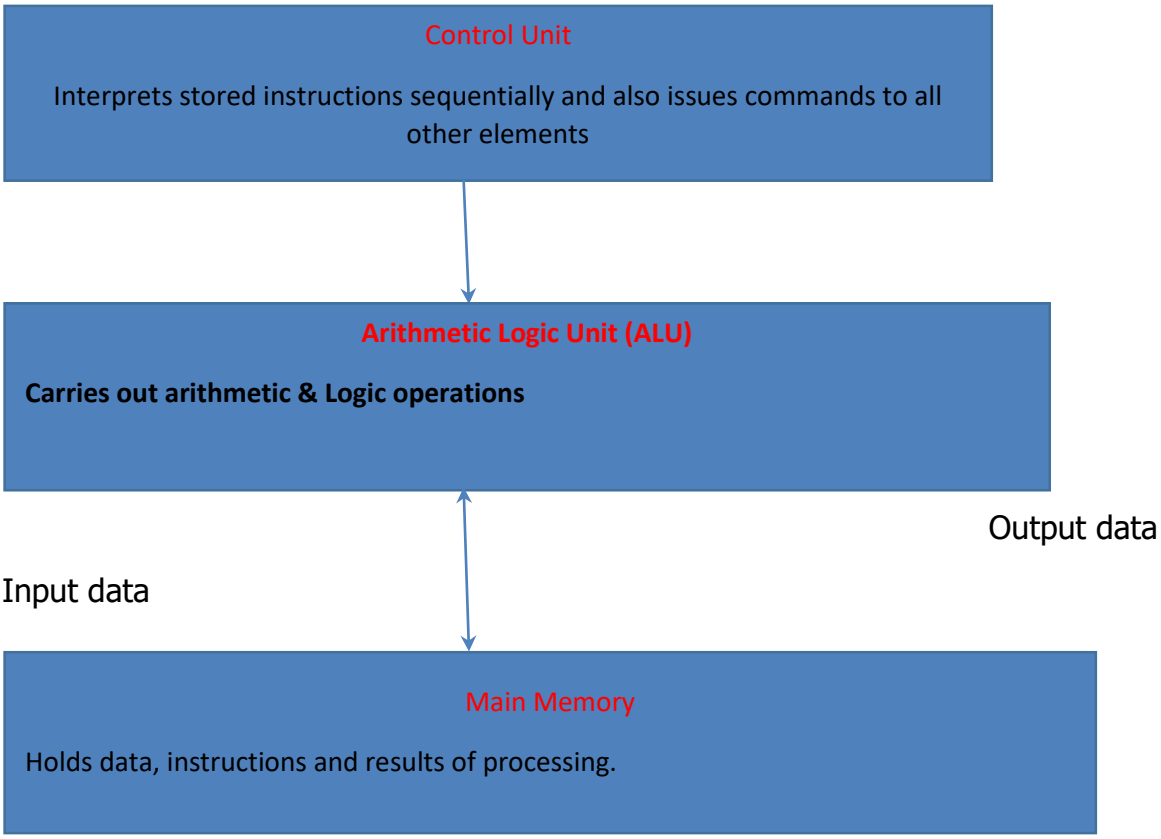
Optical mark recognition is the process of **capturing** data by shining light onto the **document.**

(c). Optical Character Readers and Optical Character Recognition (OCR)

Optical Character Reader/Recognition is a type of computer software designed to translate images of hand written or type written text (usually captured by a scanner) into machine editable text.

CENTRAL PROCESSING UNIT

- The processor, also known as the central processing unit (CPU), is the brain of the computer.



- **Memory** is the storage area. It is a facility to contain the data and program during processing.
- **Arithmetic Logic Unit (ALU):** is where calculations and logical operations take place.
- The computer must be able to do arithmetic operations on data in order to process the data.
- **Control Unit (CU):** controls the operations of the memory.
- In short it controls the movement of data.
- In modern electronic systems the **ALU** and **CU** are combined to form the **Central Processing Unit (CPU);**

OUTPUT DEVICES

- Output devices such as printer, speakers, monitors and data projectors **are used to convey information to the user.**

A printer; produces text and graphics on a physical medium such as paper.

A monitor: displays text, graphics and video on a screen

Speakers; allow you to listen to music, voice or any other audio (sounds).

A data projector; displays the text, graphics or video on the monitor on a larger screen for an audience to see clearly.

- The most commonly used output devices **are monitors, printers** and **speakers.**

STORAGE DEVICES

- **Storage devices** keep data, instructions and information permanently.
- **Storage device** refers to any hardware that is capable of storing and retrieving data.
- **Storage devices** can also be defined as anything that is capable of storing data in digital format or any hardware that is capable of holding information in digital form...
- **STORAGE:** Refers to the safe keeping of the data that has been entered into the computer so that it can be used when required.
- Storage media such as CDs, DVDs, flash drives and hard drives are used to store data, instructions and information for future use.
- For example, a CD/DVD writer is a storage device that writes to CD or DVD storage media.

APPRECIATING VARIOUS TYPES OF STORAGE MEDIA

- **Storage media** holds data, information and instructions permanently.
- The data is stored on storage media (singular: medium) such as Compact Discs (CDs), hard disks, Digital Versatile Discs/Digital Video Discs (DVDs) and Universal Serial Bus (USB) flash drives.

- **Storage media** are the physical materials on which data and information is stored.
- When you need to run a program, it is retrieved from secondary storage and loaded into the computer's primary memory, which is temporary.
- A computer reads or writes information to or from a storage media using storage devices such as CD/DVD drives or hard drives.

Secondary storage is measured in kilobytes kB. The following are used to measure large memory devices:

1024 bytes = 1 kB (kilobytes)

1024 kB = 1 MB (megabyte)

1024 MB = 1 GB (gigabytes)

1024 GB = 1 TB (terabytes)

HARD DISK

- Internal hard disks are the most popular form of storage.
- A Hard Disk **is a device that holds all the information that is stored on a computer**
- The capacity of a hard disk is measured in **gigabytes (GB) or terabytes (TB).**
- Dirt damages data stored on the hard disk, which could cause the hard disk to crash.
- **A crash** is a state that occurs when a computer or program stops functioning properly and exits after finding errors.

PORTABLE HARD DISK

- A portable hard disk is a stand- alone device that connects to a USB or fire wire port in the system unit, or wirelessly.
- It has a storage capacity of up to 4 terabytes (TB).

USB MEMORY STICK

- A USB memory stick or flash disk is a portable storage medium.
- When inserted into a USB port on a computer, you can write files to it and erase files from it.
- Flash disk are ideal for transporting files and for storing backups.
- However, they can easily be lost or misplaced because they are small.
- Their storage capacity is lower that the capacities of portable hard disks.

CDs, DVDs and Blu- ray

- Compact Discs (CDs) are an optical storage medium that uses laser technology to read and record data.
- The capacity of a CD is about 700 megabytes (MB).
- The process of writing to a CD is referred to as **Burning a disc.**
- CDs have different formats such as CD ROM, CD-R and CD-RW format.
- CD-ROM disc drive can only read CDs.
- A CD-RW disk drive can read from and write to CDs.
- Some modern CD/DVD drives can also burn a title to a CD that has been specially treated to allow this. This process is called **LightScribe.**

DVDs

- Digital Versatile Discs/Digital Video Discs (DVD) are optical storage media that use laser technology to read and record data.
- DVDs come in the following formats: DVD, DVD +RW, DVD – RAM.
- A DVD's capacity is between 4.7 GB and 17 GB, so it is suitable for storing multimedia content and at greater speeds than a CD.
- DVDs can be used to back up important information.

MEMORY CARDS

- Memory card storage is found in digital devices, such as cameras, video cameras, handheld devices, some cell phones and MP3/MP4 prayers.
- CompactFlash (CF), Secure Digital (SD) and Multimedia Card (MMC) are examples of different types of memory cards.
- The storage capacity of memory cards ranges from 8 MB to 32 GB.
- Information on a memory card can be accessed by connecting the digital device, such as a camera, to the computer via a USB cable.

TYPES OF STORAGE

There are two main storage devices namely Primary Storage / Main Storage and the Secondary Storage / Backing Storage Media. The primary storage is mainly made up of computer chips such as the RAM and ROM.

PRIMARY / MAIN STORAGE MEDIA

THE RAM

RAM stands for Random Access Memory. RAM is a volatile memory used for data storage on a computer. The memory of RAM is said to be volatile because it loses all the data stored in it when the computer is shut down.



THE ROM

ROM stands for Read Only Memory. Data stored on ROM is held even when power is switched off and therefore is considered to be non-volatile memory. **SECONDARY**

Removable and Non – Removable Disc Drives

(a). Removable Storage Devices: These are storage devices that can easily be plug to a computer and unplug after using them they can also be said to be storage media that can be ejected from the when it still running, and typical example include USB flash drives, External Hard Disc Drive, Memory Cards, Floppy Disc Drive etc.

USB Flash Drives / Memory Stick / Thumb Drives

A USB (Universal Serial Bus) flash drive is a small, portable storage device that plugs into a computer's USB port. Like a hard disc, USB flash drives stores information, but usually a lot less than most hard disc drives







COMMUNICATION DEVICES

- **A communication device** is used to send (transmit) and receive data, instructions and information between one or more computers or mobile devices.
- **A modem** is an example of a communication device.
- **A modem** is a device that converts digital information transmitted by a computer into a form that can be carried over by a regular telephone line and then re – converted it back into digital form before it is received by the second computer at the other end.

BASIC OPERATIONS

POSTURE: is the correct way of sitting when working with the computer.

- Make sure that you are 18 inches away from the monitor
- Make sure that your elbow is below the keyboard
- The legs should form an angle of 90 degrees
- Your back must be resting on an adjustable chair.
- The centre of your body should be opposite the j key.
- Your head must be upright.

COMPUTERS AND PERIPHERALS

Powering up and powering down.

- You switch a computer on by pressing the power button. When switched on, the computer automatically loads the operating system.
- The process of starting a computer is called the **booting** or **boot process**.
- When a computer is off and you switch it on, this is referred to as a **cold boot**
- When you restart a computer while the computer is on, this is referred to as a **warm boot** or a **hot boot**.

  Powering down a computer

- Click on the start button
- Click on the shutdown button.

  SECURITY AND SAFETY

  HARDWARE THEFT

- Hardware theft **refers to stealing computers and computer devices.**

CONSEQUENCES OF HARDWARE THEFT.

- **Loss of information:**
- **Inconvenience**
- **Cost**
- **Software theft**:
- **Information theft:**
- **Identify theft**

USER INTERFACE

- A user interface is a system by which people interact with a machine. The user interface includes hardware physical and software logical components.
- **COMMAND LINE INTERFACE**
- This means that the user has to communicate with the computer using typed words.
- **MENU DRIVEN INTERFACE**: a way by which a user can interact with a computer. The user can interact with a computer by choosing command and available options from a menu using a keyboard or a mouse.

- **GRAPHICAL USER INTERFACE** (GUI)
- This uses images as well as words to prompt user choice. Many graphical user interfaces are designed as **WIMP i**nterface. **WIMP** is an acronym, with the letters standing for **windows icons**, mouse or **menu** and **pointer.**
- **WISYWIG** – What You See Is What You Get

COMPUTER VIRUSES
DEFINITIONS.

- Computer virus refers to a program which damages computer systems and destroys or erases data files.
- A virus is a program that disturbs the way the computer is supposed to be running.

- **Malware is the name given to any software that could harm a computer system.**
- Malware is short for malicious software.

TYPES OF COMPUTER VIRUS

TIME BOMB.

A time bomb is a virus program that performs an activity on a particular date. Bombs hide in usual files and do not duplicate themselves.

LOGICAL BOMB

A logical bomb is a destructive program that performs an activity when a certain action has occurred.

WORM VIRUS

A worm is also a destructive program that fills a computer system with self-replicating information, clogging the system so that its operations are slowed down or stopped. Worms do not need to attach themselves to other programs. entire network. It is important to avoid foreign diskettes being used on your computer.

TROJAN HORSE

Trojan horse is a destructive program. It usually pretends as computer games or application software. A Trojan horse will hide or disguise itself as something else. It could disguise itself as part of a game or graphics demonstration program.

EFFECTS OF VIRUSES

- Delete vital files
- Windows crashes
- System operation slows down noticeably
- Hard disk controller failure
- Hard disk stops booting and files are corrupted.
- Displaying irritating messages to deleting all the files on your computer.

SOURCES OF VIRUSES

- Using other people's infected floppy disks

- Copying illegal software
- Starting the computer with infected floppy disks in the drives.
- Opening electronic mail messages or attachments which have been infected.
- Down- loading infected software from the internet.
- Exchanging flask disks

HOW CAN ONE PREVENT COMPUTER VIRUS INFECTION?

- Do not use other people's diskettes unless you first check for viruses.
- Do not illegally copy software and uses it with your machine
- Beware of internet downloads, some of them may have a virus.
- Ensure that there is no floppy disk in the drives when you boot up the computer.
- Avoid opening Spam electronic mail messages Install anti- virus software, and keep it up to date.
- Install security updates for your operating system and programs as soon as possible.

HEALTH EFFECTS OF ICT

If we use a computer for many hours (as people often do at work), there are some health issues that might affect us...

(1). Eye-Strain

One health issue that can occur after using computers for a long time is **eye-strain** (tiredness of the eyes). This is caused by looking at a **monitor** which is a **constant distance** away. The muscles that focus your eyes do not move, and so get **tired** and **painful**. Eye-strain can also cause **headaches**.

This problem can be solved:

- **Look away** from the monitor at **regular intervals** – re-focus on distant or close objects to exercise the muscles in the eye.
- **Take regular breaks**.
- Use an **anti-glare filter** in front of the monitor to cut down on screen reflections that can also tire the eyes.

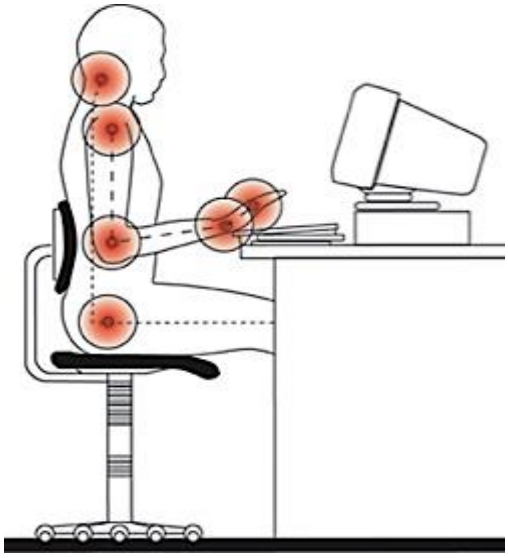


(2). Back and Neck Ache

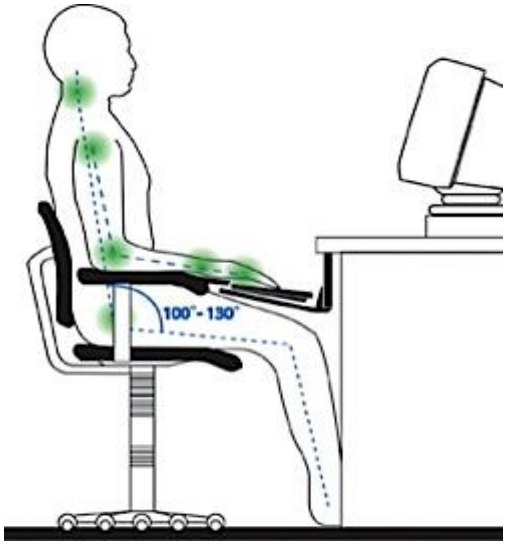
Many people suffer from **back and neck pain** after working at a computer for a long time. This is usually due to them having a **bad sitting posture**.

This problem can be solved:

- Use an **adjustable, ergonomic chair**, and take the time to s**et it up properly**.
- The computer **keyboard** and **monitor** should be at the **correct height** for the seated person (keyboard lower than the elbow, top of monitor at eye level).
- **Take regular breaks**: get up, walk around, stretch your muscles



Bad Posture



Good Posture



The science of how we interact with the objects around us is called **ergonomics**.
An **ergonomic chair** is one that **fits the body** well, giving support to areas such as the lower back (lumbar region)

(3). Repetitive Strain Injury (RSI) in Wrists and Hands

Any **repetitive** movement (same movement over and over again) can result in a health problem called **Repetitive Strain Injury** (RSI).
In particular, **typing** and using a **mouse** for long periods are common causes of **RSI** in the **wrist** (it is often called carpal-tunnel syndrome).

This problem can be solved:

- Use a **wrist-rest** to support the wrists while typing and when using the mouse.
- **Take regular breaks** from typing or using the mouse.



A rest wrist



SAFETY ISSUES WITH ICT

You wouldn't imagine that using computers could be dangerous, but there are a few situations that can result in accidents...

(1). Trailing Cables

Computer equipment is often connected to lots of **cables**: power, network, etc. If these cables are lying on the floor, they can cause people to **trip** over them.

Solution: Place cables inside cable ducts, or under the carpet / flooring



(2). Spilt Drinks or Food

If any **liquids** are **spilt** on electrical equipment, such as a computer, it can result in **damage** to the equipment, or an **electric shock** to the user.

Solution: Keep drinks and food **away** from computers, at school, do not enter the computer lab with food stuffs or drinks.



(3). Overloaded Power Sockets

Plugging **too many power cables** into a socket can result in the socket being **overloaded**, overheating, and a **fire starting**.

Solution: Never plug **too many cables** into a socket. Always make sure there are **fire extinguishers** nearby



(4). Heavy Objects Falling

Many items of computer equipment are very **heavy**: CRT monitors, laser printers, etc. Heavy items can cause serious injury if they fall on people.

Solution: Make sure equipment is placed on **strong tables** / shelves.

UNRESPONSIVE COMPUTERS

A computer is said to be unresponsive (i.e. not responding) when one of the following happens;

- ❖ When it freezes or stops loading during the booting process.
- ❖ When it freezes or stops during the process of shutting down.
- ❖ When it freezes or hangs during a process.
- The process of starting a computer is called **Booting**
- ❖ **A cold boot** is when you switch on a computer that was off
- ➢ **A warm boot** is when you restart a computer that is already on.

THE BOOTING PROCESS.

- ➢ During the booting process, the following, take place inside the CPU. First, the CPU performs a BIOS test to make sure that all computer hardware is correctly connected and functioning well. The BIOS tests also check the system clock, adapter cards and RAM chips. These tests are called POST (Power On Self – Test). Once the POST is successfully, the BIOS test checks the operating system files. Depending on the setup, the operating system files may be located on the hard disk. Alternatively, the files may be on a CD, DVD or a peripheral device like a USB.

- ➢ **SECURITY AND VIRUSES**

Usernames and passwords

Security issues associated with ICT use

- ➢ It's irresponsible to set up a computer without regard to security.
- ➢ **Authentication;** a process by which you verify that someone is who they claim they are.
- ➢ **Identification** verifies that an individual is a valid user.
- ➢ **Authentication** verifies that the user is the person he or she claims to be.
- ➢ **Logon** is the process of entering a **username** and a **password** to gain access to a computer.

- **Logoff** is the process of **disconnecting a user** from the **computer** or **network**.
- **A user account** is set up for each user, which allows him or her to log on to a computer or network.
- The user account consists of a ***username*** (user ID) and ***password.***
- ***A username*** is a series of characters that uniquely identifies a specific user.
- Every user has a username.
- **Password:** a secret combination of letters and numbers that you enter on a computer to access the system.
- **A password** is a combination of characters that is linked to the user name and is used to control access to a resource.
- When compiling a password, users need to ensure that the password is a minimum of eight characters, consists of a good mix of characters (alphabetical letters, numbers and symbols), uses uppercase and lowercase, and is not a birthdate, family name or anything else that is easy to guess.
- Passwords must be kept secret, and must be changed on a regular basis.

SAFETY PROCEDURES

Antivirus software

- **A computer virus** is a computer program specially created to cause harm on a computer system.
- Against other harmful or malicious software, also known as **malware.**

COPYRIGHT

Software product ID numbers

Licensed software and authenticity.

- **A license agreement** is a contract between the manufacturer and the user one the right to use the software.
  - **Certificate of authenticity**: this is a document, card or sticker label included with a software program that proves the customer purchased it legally.

OPEN- SOURCE VS PROPRIETY SOFTWARE

- **Proprietary software** these are software that can be bought from computer stores or downloaded from websites at a fee
- **Open source software** is usually downloaded from the internet.

Freeware, shareware and end – user licenses

- **Freeware software** software that is offered by the software copyright owner at no cost.
- **Freeware:** software that is free to use
- **Shareware** is copyright software that is distributed free for a trial period. Once trail period has expired, the user must send payment to the software developer to continue using it.

COMPUTERS IN EVERY DAY LIFE

Household, entertainment & recreation appliances

Washing machines: the microprocessor unit in the washing machine has control and sensing systems, for such things as tracking the speed of the motors, wash temperature and electrical failures and value responsible for controlling the parts that clean the clothes.

Mobile phone: the mobile's microprocessor runs all the programs installed on the phones such as the address book, estop watch and music player programs.

A Security alarm: a security alarm is a system designed to detect an authorized entry into a building or area. They are used in residential, commercial, industrial and military properties for protection against theft.

Television: the microprocessor in the television facilitates reminder of programs and sleep brightness and other controls.

Digital camera: Digital cameras also has embedded computers. When a person uses the various dials and buttons to select options, the microprocessor accepts these as inputs and performs the required actions. Some of the programs that the

Microwave ovens: the embedded microprocessor in a microwave oven is specifically designed to run the front- panel controls and all the timing and electronics of the oven.

Fridge

Digital watches: a digital clock is one that displays the time digitally.

TV, electronic game, radio

OFFICE EQUIPMENTS

Computers: this is an electronic device used to process data.

Printers: a printer is a device that produces text and graphics on a physical medium such as paper or transparency film.

Fax machines: a method of rapidly sending and receiving written material between two machines, using the telephone system.

Projector: it is a hardware device, with which an image like a computer screen is projected onto a flat screen. Projectors is often used in meetings to make presentations, because they allow for a large image to be shown.

Scanners: these are devices which can read printed or magnetic data and convert it into electronic signals to be sent to an attached computer. Scanners work, in essence, like a photocopier.

SOCIAL AND ECONOMIC EFFECTS OF COMPUTERS

- ***Security concerns***: poor security may result in physical damage, both accidental and deliberate. Poor security may result in loss of records; loss of confidentiality, sensitive information may be obtained by outsiders.
- ***Deskilling***: when the amount of skill needed to do a job is reduced. Usually because of the use of computers**. Automation is the replacement of human labour by machines.** Automation happens in many companies. Banks for example, have shed thousands of jobs as computers have replaced people.
- ***Backups:*** copies of files or data to keep in case the original files or data are lost or damaged.

- ***Firewall***: a program that accepts or rejects requests to access the computer.
- ***Head crash***: when the read or write heads damage the surface of a hard disk.
- ***Data encryption***: scrambling the data so that no one can read it without the code to unscramble it.

COMPUTER MISUSE: computers can be misused by fraudsters. **Fraud normally involves staff removing money from the company to their own pockets.**

- ***Computer crime***: using a computer to commit a crime.
- ***A cybercrime***: is a criminal act committed online.
- ***Cyberbullying***: threatening or humiliating someone online
- ***Hacking***: accessing a computer without permission to steal information or damage the software.
- ***Piracy***: the unauthorized use of somebody else's work.
- ***Source code***: the basic programming of software.

ATMS (AUTOMATIC TELLER MACHINES)

The use of ATMs or cash dispensers has removed the need to rush to the bank before it closes. ATMs work 24 hours a day, so money can be deposited or withdrawn at any time of the day. These ATMs are computer terminals connected to the central computer

What do I do to obtain cash from ATMs?

- Insert your bank card into the slot.
- Enter your PIN
- Choose the service you want, or simply press the button next to the amount you want and wait.
- Take back the bank card
- Take the money

- Take the receipt if you asked for it.

The card used to withdraw the money is plastic with a magnetic stripe similar to a credit card. The magnetic stripe is coded with the following data:

- Card holder's account number
- The PIN personal Identification Number.
- The sort code
- The card holder's withdrawal limit
- The amount withdrawn so far that day.

INFORMATION TECHNOLOGY IN EVERYDAY LIFE

Connect a standard decoder to a TV

Connect a DVD player to a TV

Connect a hi- fi to a TV

Connect an LCD projector to a computer

Scanners

Printers

Digital cameras

Cell phones

How to connect a cell phone to a computer

Social networks

- An online social network, also called **a social networking** site, is a website where users set up their personal online profiles and build social groups with other users based on common interests.

COMPUTER BASED LEARNING

- Refers to the use of computers to aid and support the learning process.
- **E – learning:** distance learning using the internet.
- **Video Conferencing:** using sound and video technology to communicate over the internet

- **Digital presentation:** an electronic presentation of information using text, images sound and video.

COMPUTERS IN BANKING AND E- COMMERCE

Using computers to organise, retrieve and process data

FILES

- A computer file is an item that contains information and is stored on a storage medium, such as a hard drive, solid- state storage, CD or DVD.
- Information such as text, images, music and videos can be stored in a file.

FOLDERS

- A folder is a logical representation of a location where you can store files or other folders.
- A folder is based on the actual folders use to store documents in a real-life filling cabinet
- A folder within another folder is referred to as a **subfolder**

DIRECTORY

- A directory is a file system classification that contains a set of computers files and other folders.

- **LIBRARIES**
- A library contains a set of links to similar files that are stored on various devices and in various folder.
- A library is similar to a folder. It does not actually store files, rather it provides a view of files.

ELECTRONIC FUND TRANSFER

- Is about making payments for goods and services using computers. Money is transferred from the bank account of the person buying goods and services and paid into the account of the person selling.

NETWORK FOR ELECTRONIC TRANSFER (NETS)

- Was developed for use in Singapore and is now also in use in the Philippines, South Korea and Thailand.
- This system enables bank customers to withdraw funds from their banking accounts at retail outlets using ATM cards.
- The aim of NETS is to improve accessibility to cash withdraw facilities.

GIRO is a payment transfer system from one bank account to another bank account and is initiated by the payer.

COMPUTERS IN BANKING AND E- COMMERCE.

- This is the buying and selling of goods and services through the internet.
- It is also called as **on-line shopping.**

Point of sale

Common uses of computers in supermarkets

- When a person pays for goods and services and the data is captured at the same location, this is known as the **Point of Sale (POS).**
- **Barcoding:** the process of identifying the goods.
- **A Barcode:** is a pattern of lines on a product label. This pattern of lines are usually read by means of a scanners which are powered with laser beams.

NETWORKS AND THE INTERNET.

- **A computer network**, simply referred to as a network, is a group of two or more computers and devices linked together so that they can communicate with one another over a shared network medium.
- The sharing resources over a network is called **networking**.
- Information and devices for users on a network is called **shared resources**
- When a computer is connected to a network, often wirelessly, it is **online**.
- The world's largest computer network is the **internet**.

✓ **The computer network** is an interconnection of computers and other related devices using media links.

✓ **A network** is an interconnection of computers.

- ✓ **A computer network** is any grouping of computers which are connected to each other and are able to exchange data.
- ✓ **Radio:** waves of sound that carry information.
- ✓ **Wired networks:** a network connection made with physical cables.
- ✓ **Router:** a switch that can connect computer devices to a telephone line.
- ✓ **Switch:** a network device that connects all the computers and devices on a network.
- ✓ **Wireless devices:** have no data cables. They send signals through the air using radio waves, infrared light or microwave.
- ✓ **A wireless network** also has a switch to connect the devices on a network.
- ✓ A wireless switch is called a **wireless access point.**
- ✓ **Network coverage:** the geographic area in which a computer network operates.

TERMINLOGIES ON NETWORK

NODE: a device that is attached to a network.

SERVER: a computer shared by several users in a network.

HOST: a host computer- the main computer that controls a network.

CLIENT: a requesting computer in a network.

MEDIA: are the collective communication outlets or tools that are used to store and deliver information or data.

HACKER: these are people who gain unauthorised access to computers or telecommunication systems for the challenge of it.

- ➢ **COMPUTER HACKERS** are people who are mainly self- taught computer specialists or hobbyists.

- **Hacking is when an unauthorised user gains access to a computer system or files to which the unauthorised user is not entitled to have access**.

CRACKER: they gain unauthorised access to information technology but do so for malicious purposes.

MEDIA DEVICES/NETWORK DEVICES

MODEMS: A MODEM is a device that converts the digital data from the computer into an analogue from that is suitable for transmission by conventional telephone systems.

HUBS: are network hardware devices which are used as a central connecting point for joining transmission lines in a star network.

BRIDGE: bridges are network hardware devices which are used to connect network segments or networks.

ROUTERS: routers are network hardware devices that are used to connect two or more network segments or sub networks together.

BROUTERS. Brouters are network hardware devices which combine the functions of bridges and routers. The term brouter is made out of the two Words Bridge and router.

GATEWAYS. A gateway is a method of enabling communications between two or more network segments.

PRINT SERVER. A print server is a node on a network. It shares its printer facilities among all its users on the network.

ADVANTAGES OF NETWORKS

- ✓ Sharing of data and programs.
- ✓ Sharing of peripherals e.g. printers.
- ✓ Improved communication facilities.
- ✓ Easy distribution of work and processing loads.

DISADVANTAGES OF NETWORKS.

- ✓ Virus transmission easy.
- ✓ Relatively expensive to set up.
- ✓ The system is prone to hackers, crackers and eavesdropper.
- ✓ Need for power energy to work.

Differences between World Wide Web and the internet.

- World Wide Web is also called a web. While the internet is also called the global village.
- World Wide Web can be used to view or transfer documents across the internet. Whereas electronic mail and file transfer are some of the services the internet offers.
- The World Wide Web hosts a massive amount of information scattered around the globe. While the internet allows people from different parts of the world to communicate with each other as though they were just in the same building.

A LOCAL NETWORK: is a privately-owned network that serves users within a confined geographical area.

TYPES OF LOCAL NETWORKS

LOCAL AREA NETWORK

- **A local area network** is a computer network that interconnects computers within a limited area such as a home, school, computer laboratory, or office building, using network media.
- **Local area network (LAN)** a network that covers a small area, such as a room or a building.
- Each computer or device in the LAN is known as a **node.**

CLIENT/ SERVER NETWORK

- Client/ server LAN consists of requesting microcomputers called client, and supplying devices that provide a service, called server. E.g. SAN.
- **PEER- TO- PEER LAN:** is one in which microcomputers on the network directly communicate with one another without relying on a server. E.g. PAN

PERSONAL AREA NETWORK (PAN) AND HOME AREA NETWORK (HAN)

- A personal Area Network is computer network organised around an individual personal device or covers and individual's device.
- These devices are connected within a user's personal space within an area the size of a desktop.

A HOME AREA NETWORK (HAN) spans a larger area within a house or home office and is typically used by more than one person.

- The internet is the world's largest WAN.

METROPOLITAN AREA NETWORK (MAN)

- is a network covering a geographical area the size of a city, country?
- It is designed to service a metropolitan area such as a city, a town or an entire country.
- Wireless MANs are often referred to as **municipal Wi – Fi projects.**

STORAGE AREA NETWORK (SAN)

- **A Storage Area Network (SAN)** is a high- speed network that connects a collection of computers or servers to storage devices using optical cables.
- It is sometimes referred to as the **network behind the servers.**

WIDE AREA NETWORK: is a network that covers a wide geographical area such as a province or state or the entire country.

Wide area network (WAN) a network that covers a large area, such as a city or country.

- WANs transmit data by telephone links or satellite links over very long distances, e.g. banks network of ATMs.

- The Internet is the world's largest WAN.
- **GLOBAL AREA NETWORK**: is a type of computer network linking the whole world. It is often referred to as the internet.
- **Internal network**: a network that belongs to an organisation.
- **A server**: is connected to all the other computers on the network.

DIFFERENCE BETWEEN LAN, MAN and WAN

- Local area networks usually cover one floor of a building, the whole building, a campus or a factory area. A metropolitan area network may cover a whole town while wide area network may cover a whole district, province, country, continent or even the whole globe, as is the case with the internet.

DIFFERENCES BETWEEN INTRANET AND EXTRANET

- An intranet is a self- contained system of company information. Whereas an extranet is that company network beyond the intranet.
- An intranet may not have a physical connection with the internet, while extranet have physical connection with the internet.

THE INTERNET

- The internet is a network of networks which interconnects several networks on our globe today.
- **The internet** is a network that covers all the countries in the world.
- It is made up of computers, devices and communication media.
- The computer is connected to a special switch called **a router.**
- **The router** connects to an Internet Service Provider (ISP) over a telephone line.
- **The internet** is a worldwide network of computer networks.
- **World wide web (www):** the software that runs on the internet.
- **The world wide web (www) it** is called the **Web.**

- It is an interconnection of large and small networks around the globe.
- The Internet began in 1962 as a resilient computer network for the US military and over time has grown into a global communication tool of thousands of computer networks and millions of computers that share a common addressing scheme.
- The internet is the global network that allows people from different parts of the world to communicate with each other as though they were just in different rooms of the same building.

GLOBAL VILLAGE

- The global village refers to the situation that has been brought about by the marvellous strides taken by developments in the chip technology and telecommunications.
- Today one can attend a lecture held in New York while in his room in kabulonga, Lusaka, Zambia. This is no longer a dream but a reality.
- The world has become as small as a village - the global village. The physical distance that separates the people of this planet is no longer intimidating as electronics has made the distance virtually smaller.

USES OF INTERNET

- Online shopping
- Advertisement
- Online Baking
- Online insurance
- Online travel and booking
- Watching Videos/Downloading/listening to music

E-learning/research, instant messaging

UNIVERSAL RESOURCE LOCATOR

- This is just another name for a web address.
- The URL consists of the name of the protocol. (Usually HTTP or FTP) followed by the address of the computer you want to connect to.
- Each page on the Internet has a unique web address called a URL.
- The address you enter is called a universal resource locator, or URL for short.

- Every website has a URL, and no two URLs are the same.
- **Universal resource locator (URL):** the address of a website.
- **The home page** is the starting page in a web browser or on a website.
- **Home page:** the web page that appears each time you open your web browser or a website.

ELECTRONIC MAIL

- Electronic mail is a way of exchanging data/information using electronic transmission.
- The e- mail facility has contributed much to the reduction of paper work in the automated office.
- The e- mail facility requires a computer network or an interconnection of computers networks.

SEND AN EMAIL

- Step 1: log in to your Gmail account.
- Step 2: Click the Compose button
- Step 3: in the space next to "To", type the other person's email address.
- Step 4: in the space next to "subject", type a few words to say what the email is about.
- Step 5: write (or compose) the email.
- Step 6: Click send.

EMAIL SCREEN LAYOUT

- An inbox for emails you receive
- A Drafts box: for emails you have written but have not yet sent
- A Sent box: for emails you have sent
- An Outbox: for emails you have written and will be sent when you go online.
- A Trash can: for emails you want to delete.

WI- FI

- Is a wireless network technology used for connecting computers to a wireless network or to the internet?
- It is a form of low-power wireless communication used by many electronic devices such as laptops, systems, smart phones, etc.

- In a Wi-Fi setup, a wireless router serves as the communication hub.
- **A hotspot** is a wireless network that provides internet connection to portable computers and other devices. The wireless connection is provided by a wireless router,
- **Cloud computing:** IT companies on the Internet who change customers to use their servers and storage media.

Modem

- A dial- up modem **converts digital signals to analogue signals to digital signals so that data can travel on analogue telephone lines.**
- A digital modem uses a digital line such as ADSL, which is a permanent connection.

Router

- A router is used to connect a computer on a network to the internet.

ADSL

- Asymmetric Digital Subscriber Line (ADSL) is a high- speed internet connection using regular telephone lines.

INTERNET SERVICE PROVIDER (ISP) AND ISP INTERNET SERVICE.

- An ISP is a company that provides connection to the internet, and can be a regional or national ISP.
- **Internet service provider (ISP):** a company that provides Internet access to users.
- Examples of ISPs: **ZAMNET, MTN Business, Airtel or I way Africa.**

WEB BROWSER.

- **A web browser** is a program that enables you to find, retrieve, view and send documents over the Internet.
- The web browser that comes with the windows operating system is **Internet Explorer.**

- The most commonly used web browsers **are Microsoft Internet Explorer, Mozilla Firefox, Google Chrome and Opera Mini for Mobile, amongst others.**

THE WORLD WIDE WEB

- The World Wide Web is a service offered on the internet
- A collection of web pages is known as a **website.**
- For users to retrieve, send, store, and display media on websites, a special software application is needed, called a **web browser**.

SEARCH ENGINES

- **A search engine** is a website or application that enables users to find information quickly and easily by searching documents.
- **Search engines** help you find all sorts of information.
- **Databases:** collections of files that contain information grouped into categories.
- **Search terms:** the word or words that describe the information you want to find.
- **Some of the best – known search engines are Google, Yahoo, Excite, Alta Vista, Ask and Bing.**

DIFFERENCE BETWEEN WIRED AND WIRELESS NETWORKS

- Wired network is an interconnection of computers which reside in one computers or networks which reside in areas that are separated by large physical distances.
- Wired could be a single room, several rooms on one floor of a building or on each floor of a building. Whereas wireless can connect computers in different towns, provinces, countries, continents or the entire globe.

NETWORK TOPOLOGIES

- **A topology** is the geometric arrangement of a computer system.
- **A Star network** ties end user computer to a central computer as shown.
- A message sent to another computer goes through the hub or switch which in turn interprets the address and sends the communication to the right compute.
- **A Bus network** is a network in which local processors share the same bus, or communications channels.
- A message sent to another computer travels the entire bus along with a unique address for the destination computer.

- **A ring** network, on the other, ties local computer processors together in a ring on an equal basis forming a closed loop.
- As with the bus topology, all communications travel the entire communications medium and each computer must read the address and determine if it is the intended recipient.
- **Wireless Communication**: the term wireless refers to the communication or transmission of information over a distance without requiring wires.
- The communication is set and the information is transmitted **through the air**, **without requiring any cables**, by using electromagnetic waves like radio.
- **Bluetooth technology**: allows you to connect a variety of different electronic devices wirelessly to a system for the transfer and sharing of data and this is the main function of Bluetooth.
- **Encryption**: a way of coding the information in a file or e-mail message so that if it is intercepted by a third party as it travels over a network it cannot be read.
- **Hyperlink**: a link in a document to information within that document or another document.
- **Cache**: the location where visited web pages are stored, enabling them to be displayed more rapidly at the next visit.
- **File transfer protocol**: permits users to transfer file from servers to client and vice versa.
- **Packet**: is a discrete unit into which digital messages are sliced for transmission over the internet.
- **Packet switching**: a method of slicing digital messages into packets, sending the packets along different communication paths as they become available, and then reassembling the packets once they arrive at their destination.

- **IP Address**: millions of computers are able to communicate to each other because each computer on the internet has an address which makes it possible for it to send and receive TCP packets.
- This address is known as the **IP address (Internet Protocol address)**
- **Internet Protocols:** is a set rules and standards for data transfer which governs the formatting, ordering, compressing, and error checking of messages.
- **File Transfer protocol:** one of the original Internet services as a way of transferring data from one place to another over the Internet.
- **The protocol** is often used for downloading large files from a website.
- **Hypertext Transfer Protocol:** this is the Internet protocol used for transferring web pages.
- **Simple Mail Transfer Protocol:** is the internet protocol used to send mail to a server.
- **Secure socket layer (SSL):** this is a protocol that provides secure communication between a client and a server computer.
- **Hypertext Markup Language (HTML):** is the language used to create web pages on the Internet.

MULTIMEDIA

What is multimedia?
The term multimedia is coined from two terms: multiple and media. Hence multimedia means using multiple Medias to communicate.

Multimedia is the combination of different media elements like text, audio, graphics, video and animations.

- Text, data, image and graphics media are known as **discrete media**.
- Audio and video media are known as **continuous media**.
- **Interactive multimedia** is the means to interface/communicate with computer keyboard, mouse, touch screen, on screen buttons, and text entry.

  ELEMENTS OF MULTIMEDIA?
- Text, Video, Audio, Animation, Graphics,

  TEXT: Words treated as data by a computer.
- **Hypertext** is a text which contains links to other texts.
- **A typeface** is a family of characters of letters which have similar looks. E.g. Arial-12points, Times New Roman- 14points, Algerian- 18points etc.

  EXAMPLES OF TEXT
- Arial
- Times New Romans
- ALGERIAN
- Impact

VIDEO

- The term video refers to several storage formats for moving pictures e.g. digital video formats, including Blu- ray Disc, DVD, QuickTime, and MPEG- 4; and analog videotapes, including VHS and Betamax.
- This is the Technology of electronically capturing, recording, processing, storing, and transmitting a sequence of still images in motion.

DIGITAL VIDEO: composed of a series of still image frames and produces the illusion of movement.

ANIMATION: is a visual change over time.

GRAPHICS: graphics are visual presentations on some surface, such as a wall, canvas, computer screen. E.g. photographs, drawings, Line Art, graphs, etc.

- This is an image that is generated by a computer.

IMAGES.

- An image is an artefact that has a similar appearance to some subjects.
- A **pixel** (**short for picture element**) is the small dot on the screen.
- ***A DIGITAL IMAGE***: is a representation of a two-dimensional image using ones and zeros (binary).

CLIPART

- This is art or an image that is made by various artists to fit different categories such as people, animals, school, etc. that can be inserted into your own document.

AUDIO

- In computers, audio is the sound system that comes with or can be added to a computer.
- An audio file is a record of captured sound that can be played back.

TYPES OF MULTIMEDIA DEVICES

- Microphones, speakers, Digital Cameras, headphones, Computers, DVD player, LCD projector, Video Games, Home theatres, TV.

FEATURES OF A MULTIMEDIA SYSTEM

- Very High Processing Power
- Multimedia Capable File System
- Data Representations/File Formats that support multimedia.
- Efficient and High Input/output devices
- Special Operating Systems
- Storage and memory.

COMPONENTS OF A MULTIMEDIA SYSTEM

- ***Capture devices***
  - Video Camera, Video Recorder,

- Audio Microphone, Keyboard,
- Mice, graphics tablets,
- 3D input devices, tactile sensors,
- VR devices
- Digitising/ Sampling Hardware.

STORAGE DEVICES

- ✓ Hard disks, CD-ROMS,
- ✓ Zip drives, Flash drives,
- ✓ DVD, etc.

COMMUNICATION NETWORKS

- Intranets, internets, Ethernet.

COMPUTER SYSTEMS

- Multimedia Desktop machines, Workstations, MPEG/VIDEO/DSP Hardware.

DISPLAY DEVICES

- CD- quality speakers, HDTV, SVGA,
- LCD, TFT, LED monitors, colour printers etc.

MERITS OF MULTIMEDIA

- ➢ Increases learning effectiveness.
- ➢ It improves personal communications.
- ➢ Is easy to use.
- ➢ Reduces training costs.
- ➢ It provides attention to students
- ➢ Multimedia is Entertaining as Well as Educational.

DEMERITS OF MULTIMEDIA.

- ➢ It takes time to create and compile.
- ➢ The hardware and software are expensive.
- ➢ Not always compatible
- ➢ Not always easy to configure.

STAGES IN MULTIMEDIA PRODUCTION

- Research and analysis
- Scripting or flowcharting
- Story boarding
- Collection of media elements and construction
- Programming
- Testing

USAGE OF MULTIMEDIA

- Advertisements
- Entertainment
- Design in Architects and engineers' projects
- Animations and movies
- Education and training
- Recording or broadcasting
- Using video conferencing
- Demonstrating surgeries.

APPLICATIONS OF MULTIMEDIA

- Home shopping, games, virtual reality, digital video editing and production systems
- Multimedia database systems.

COMPUTER STUDIES GRADE 9 NOTES

Terminologies used in computers

- A computer is a programmable electronic device that accepts processes and stores data, and produces output in an appropriate format according to the set of instructions or programs.

- Instructions are sets of steps that indicate to the computer how a particular task is to be accomplished.

General purpose computers

- Is designed to perform many different tasks for everyday use. It is used for most of the things that most people want to do, such as playing games, word processing, analysing numerical data using a spreadsheet program, sending and receiving email, and accessing the internet. It is generally used in homes and offices. Most computers that you see advertised are general- purpose computers.

Dedicated computers

- Is designed to perform only one specialised function to optimise efficiency or convenience.

Data and Information

Data is a collection of raw or unprocessed facts that do not have meaning of their own. Examples of data: text, numbers, images, audio and video.

Information is processed data that conveys meaning and is useful to the user or organisation for making decisions or solving problems. Data is processed into information in a systematic way through a sequence of information.

Hardware

Refers to the physical components of a computer that you can see and touch, such as the keyboard, mouse, printer, scanner and monitor.

Software

Is a set of instructions that tells the computer and its devices what to do? Hardware cannot function without software. Examples of software are Windows Media Player,

Peripherals devices

Input, output, storage and communication devices that can be connected to a computer's system unit are also called **peripherals.**

Peripherals are controlled by the processor in the computer as well as system software.

EXAMPLES OF PERIPHERALS

Modems, external hard drives and disk drives, printer's scanners and keyboards.

Personal Digital Assistant (PDA)

- A PDA is a small pocket computer with limited capabilities used to supplement and synchronise with a desktop computer.
- A PDA is used for scheduling appointments, storing addresses and phone numbers, playing games or connecting wirelessly to the internet.
- Some PDAs enable the user to make telephone calls. Most PDAs have touch screens, instead of keyboards, which you use with your finger or a stylus, which is a pen shaped pointing tool.
- Some PDAs have cameras that can be used to capture and view photographs.

Megahertz (MHz) and gigahertz (GHz)

- The processor, also called the central processing unit (CPU), is responsible for all processing that takes place in the computer.
- A system clock controls the timing of all computer operations in the CPU. **Clock speed** is measured in the number of clock pulse (or cycles) per second.
- Megahertz (MHz) and gigahertz (GHz) are the units of measurement for clock speed. **Hertz** (Hz) means cycles per second.

Bits, bytes, megabytes, gigabytes and terabytes.

- Computers use only two digits (0 and 1) to process and store information.
- We refer to these digits as bits (short for binary digit). A **bit** is the smallest unit of data that a computer can process.
- A **byte** is the basic storage unit in memory. A group of **eight bits** is referred to as a **byte.**

MEASURING MEMORY.

- A bit is in reality a Binary Digit.
- A bit is the smallest unit of data on the computer.
- One byte represents one character.
- Therefore 8 bits represent 1 byte of storage.
- Byte is the smallest usable unit of measure for memory.
- Memory is measured in ***bytes.***
- A byte is a grouping of 8 bits which the computer can operate on as a single unit.
- 1000bytes is 1 kilobyte: one kilobyte is 1024 characters
- 1000 kilobytes is equal to 1megabytes.
- 1000megabytes = 1 gigabytes 1GB.

- 1,000 gigabytes = 1 terabytes.

	KILOBYTE	***MEGABYTE***	***GIGABYTE***	***TERABYTE***
BYTES	*1,000 (1thousand)*	*1,000,000 (1 million)*	*1,000,000,000 (1 billion)*	*1,000,000,000,000 (1 trillion)*
KILOBYTES	*1*	*1,000 (1thousand)*	*1,000,000 (1million)*	*1,000,000,000 (1billion)*
MEGABYTES	*0.001*	*1*	*1,000 (1thousand)*	*1,000,000 (1million)*
GIGABYTES			*1*	*1000*
TERABYTES				*1*

Peripheral devices

Connecting peripheral devices

Ports and connectors

- Ports and connectors enable communication between peripheral devices and the system unit.
- **A port** enables a peripheral to attach to the system unit.
- A Connector attaches a cable to a peripheral. One end of the cable attaches a connector to a port on the system unit and the other end attaches a connector to the peripheral.

- The first thing to remember is the difference between male and female connectors**. Male connectors have exposed pins and female connectors +have matching holes for the pins of the corresponding male connector to fit into.**
- **When you join a connector to a port, the connector an-d port must have the same shape and the same number of pins or holes.**
- Make sure that the computer is off before inserting connectors into any of the ports, otherwise small power glitches can corrupt an open file, cause a program to freeze or cause permanent damage to the motherboard.
- A power glitch is a momentary or unexpected interruption or surge in electricity voltage.
- Some ports are hot- swappable. **This means that you can plug and unplug a device into and from a port while the computer is on.**

Common port types

Universal Serial Bus (USB) ports

- A **Universal Serial Bus** port is a high- speed connection to the computer, which enables the transfer of data between devices.
- Devices such as scanners, keyboards, mice, joystick, printers, modems and some CD- ROMs are connected to the computer using a USB port.
- A USB port is also used to supply power to devices that do not have their own power supply.
- USB ports are completely hot- swappable and plug- and – play.

Fire Wire ports

- a fire wire port is in fact a USB port, only that it is a much faster connection. It is used to connect devices that require fast data transmission such as high-speed disk drives, digital camera and digital video cameras.
- A Fire Wire port also known as IEEE 12394 is a faster connection than a USB port

Network ports

- A network port enables a computer to connect to other computers on a network. These connections allow computers to communicate with servers, as well as other computers on the network, and to share resources.
- A network port is a port which enables a computer to connect to other computers on a network, allowing computers to communicate with servers and other computers on the same network in order to share resources.

DVI, HDMI and Display Port Ports

A DVI (Digital Video Interface) port, HDML (High Definition Multimedia Interface) port and Display Port port are designed for audio visual devices such as monitors for desktop computers and for connecting an external monitor to a laptop computer.

Blue Tooth ports

Many computers, peripherals, smartphones, PDAs and other electronic devices are Bluetooth enabled. This means that they contain a small chip that enables them to communicate with other Bluetooth- enabled devices.

Thunderbolt ports

- **A** thunderbolt port is a very high – speed port used to connect display and storage devices.
- A thunderbolt port is a very high- speed port for connecting display or storage devices. Data transfers for backup, sharing and editing are faster with thunderbolt technology.

DEVICES THAT ARE BOTH INPUT AND OUTPUT DEVICES

In grade 8, you learnt about input and output devices. Input devices are used to enter data into a computer system. Some of examples of input

devices are a keyboard, mouse, touch screen, microphone, scanner, digital camera, cell phone, webcam or video camera.

Output devices convey information to the user and must be presented in the required format for the user. Output devices such as printers, speakers, monitors and data projectors are used to convey information. Devices such as touch screen, function as both input and output devices.

TOUCH SCREEN

A touch screen is a touch- sensitive display device. Users interact with the device using their fingers or a stylus. You do not manually type large amounts of data. Instead, you touch on a word, picture, number or location identified on the screen. Many ATMs and self- check- in kiosks at airports have touch screens. Fast retail businesses and restaurants use touch screens for quick and easy input.

DISK DRIVES

A disk drive also serves as both an input device and an output device. You can retrieve information from storage media such as a hard disk, CDs or DVDs using the appropriate disk drive. You can also write information to storage media using a disk drive.

Examples of disk drives are hard disks and CD/DVD drives.

AUDIO AND SOUND CARDS

An audio/ sound card is an adapter card found on the computer's motherboard. It boosts the sound- producing capacity of the computer by allowing you to plug a microphone to accept sound input. It also enables you to output sound through external speakers or a headset.

MODEMS

A modem is one of the most commonly used communication devices. Home users use a dial-up or digital modem whereas small office users use a digital modem.

A dial up modem converts digital signals to analogue signals and analogue signals to digital signal so that data can travel on an ordinary analogue telephone line.

FAX MACHINES

Although the role of a fax machine (facsimile machine) has decreased in the office with the introduction of e-mail and scanners, organisations still use either a dedicated fax machine or multifunction printer with a fax feature to send and receive faxes via telephone lines.

NETWORK CARDS

- ✓ **A** network card enables a computer or device to connect to a network.
- ✓ In a wired network, a network interface card (NIC) or network card enables a computer or device to connect to the network.
- ✓ A NIC is an adapter card that slots into an available slot in the motherboard.
- ✓ The adapter card has a port to which a network cable attach.
- ✓ A wireless network card enables wireless connectivity. These adapter cards often have a small antenna that communicate through the air with the server's antenna using radio waves.

WORD PROCESSING NOTES

WORD PROCESSING: programs are used to create documents such as letters, contracts, projects and other assignments.

Word processing: creating, editing, formatting, and printing of documents.

- **Log on:** to go through the procedures to start up and begin using a computer.
-

FIRST LOOKS

- **Recycle bin**: deleted files go here, where you can view them, get them back or finally delete them.
- **Administrator**: a user account that can control all the settings on the computer, and create other users and set their security levels.
- **Icons:** are small pictures on the desktop that launch programs or open files when you click on them.
- **Shortcut**: a special kind of icon with a curved arrow in the bottom left corner that links to applications or files for easier access.
- **Start button**: opens a menu of all applications and utilities installed on the computer.
- **Windows Explorer**: find and manage files on your computer
- **Task bar**: shows icons for all applications and documents open on the computer.
- **System tray**: shows the date and time as well as icons to access programs running in the background on the computer.
- **Side bar**: area on the desktop where a variety of gadgets can be added.

OPENING THE APPLICATIONS INSTALLED ON YOUR COMPUTER.

- **The start button:** on the bottom left of the desktop screen enables you to view all the programs installed on your computer.
- When you click on **the start button** you will see the **start menu.**
- **The start menu** lists all the available programs on your computer.

HOW TO OPEN MICROSOFT WORD

- ❖ Click on the start button
- ❖ Click on the start menu to open the programs
  OR
- Double click on an existing file and it will open in the application that was used to create it.
- Click on a short cut on the task bar or the desktop to open a program
- Type the name of the application in the search box on the start menu and click on the name if it comes up in the search results.

CLOSE AN APPLICATION

- ▪ Click on the cross in+ the top right- hand corner of the window.
- ▪ If your document or file has not been saved, the program will ask you if you want to save.

USE THE KEYBOARD IN AN APPLICATION

- ✓ Press the Caps Lock key to switch between **upper – case and lower-case letters.**
- ✓ **Upper – case** means Capital Letters
- ✓ **Lower- case** means small letters.
- ✓ This option is used only when you intend typing several letters in upper case. Press the Caps Lock key a second time to switch back to lower case.
- ✓ When in upper case mode an indicator light will be lit up on your keyboard.
- ✓ Hold the **Shift key down** while typing to insert a single upper-case letter such as at the start of a sentence.
- ✓ It is also used to type **the characters that are shown above the numbers keys such as !,?, @ % # & $.**

- ✓ Use the **Ctrl** (Control) key **in combination with other keys to perform certain actions.**
- ✓ These actions are often called **shortcuts.**
- ✓ For example, pressing the **Ctrl key** at the same time as the letter **'s' will save a document.**
- ✓ **The Tab key** moves the cursor a set number of spaces.
- ✓ **The Enter Key:** moves the cursor to the start of a new line.

MICROSOFT WORD ENVIRONMENT

- **Insertion point** – the blinking cursor where you will start typing.
- **Insertion point**: a cursor that shows you where to type in a document.
- **Plain text**: text that has not been formatted.
- **Alignment**: how the text lines up on a page.
- **Characters**: the letters, numbers and symbols in a piece of text.
- **Font**: the size and shape of the characters in a piece of text.
- **Margin**: the border at the left or right of the workspace that shows you where you can type.
- **Proof reading**: to check and correct spelling mistakes in a document.
- **Thesaurus**: a dictionary of synonyms or related words.
- **Preview**: to show printed pages as they will look on paper.
- **Drag handles**: tools that let you resize an image or shape.
- **The Title Bar:** is found to the right of the Quick Access Toolbar. It displays the name of the document on which you are currently working.
- **Zoom Tool**: used to get a better overall view of your document.
- **Status Bar**: is along the bottom of the window. It provides standard information, such as the number of pages and a word count of the document.

- **The Ribbon**: is a feature that you will find throughout the modern Microsoft Office applications. It contains a number of tabs that each contain related groups of options.
- **A dialogue box** is a pop-up box that has more detailed options.
- **Insert mode:** this means that if you place the insertion point between two words and start typing, the text to the will move along and the new text will be inserted at that point.
- **Delete Key:** When this key is pressed the character to the right of the insertion point is deleted.
- **Backspace Key:** when this key is pressed, the character to the left of the insertion point is deleted.

SAVE A NEW DOCUMENT

- Click on the Save button on the Quick Access Toolbar.
- **A dialogue box** will come up where you can choose the location in which to save your file and give it a name.
- Click on the File tab and choose the Save option.
- Use the Ctrl + S keys together as a shortcut to save the document.
- Once a document has been saved the first time and given a name, you can use the options above without the dialogue box opening.
- If you want to save a file under a different name or in a new location, click on the File menu and choose the Save As options.

OPEN AN EXISTING DOCUMENT

- To open and existing document when you already have Microsoft Word open, you need to do the following:
    - Click on the File tab.
    - then click on the Open option. A dialogue box will open.

- Select the file you want to open and the click on the Open button.

EDIT TEXT: it is possible to edit a document by deleting sections of text, making a copy of a section of text, or even moving a block of text to another place in the document.

DELETE TEXT:

- highlight a word or larger piece of text with the mouse.
- Press Delete to delete that whole piece of text.

COPY AND PASTE TEXT

- When you copy a section of text or an object, such as a picture, in an application the copy is placed in a temporary storage place in the memory of the computer.
- This storage place is known as the **Clipboard.**
- **Clipboard:** a memory space that holds a copy of text or graphics that has been copied or cut.

To copy text.

- Click on the copy button in the clipboard group on the left of the Home tab,
- Right – click on the selected text and choose Copy from the pop – up menu.
- Use the keyboard instead of the mouse by pressing the Ctrl + C keys at the same time.

The selected text or object has now been saved to the Clipboard.

To paste text

Position the insertion point where you want the copy of the text to appear in your document, then:

- ✓ Click on the Paste button in the Clipboard group on the left of the Home tab.

- ✓ Right – click at the new position and choose Paste from the pop-up menu,
- ✓ Use the keyboard instead of the mouse by pressing the Ctrl + V keys at the same time.
- ❖ **When using Copy and Paste**, the original piece of text remains in its original position.
- ❖ Sometimes you want to move a section of text.
- ❖ Then you need to use the Cut and Paste options.
- ❖ Here the text is removed from its original position when you click on the Cut option and is placed on the Clipboard.

Cut and Paste Text

Select the text that you want to move, then:

- Click on the cut button in the clipboard group on the left of the Home tab.
- Right – click on the selected text and choose Cut from the pop – up menu
- Use the keyboard instead of the mouse by pressing on the Ctrl + X keys at the same time.
- Then click on the Paste button in the Clipboard group on the Home tab.

The Undo and Redo buttons

- ➢ The Undo button is very useful if you makes a mistake.
- ➢ It will undo your recent keyboard actions.
- ➢ As soon as you have used Undo, the Redo button becomes available in case you want to reverse your Undo action.
- ➢ **Typeface:** the actual appearance of a particular font.

Format text:

- ➢ The word font refers to the specific typeface being used as well as the size of the letters.

- **The typeface** is the actual appearance of the letter.
- The size of the letters is commonly measured in Points, abbreviated to **pt.**
- Documents are commonly typed in an 11 - pt or 12-pt font.

You can change the appearance of a font as follows:

- Select the Font drop – down list to select a completely different font.
- Hold the mouse cursor over the various buttons in the Font group to see their names.
- **Font size:** the larger the number the larger the size of the letters.
- **Grow font;** increases the size of the font.
- **Shrink font:** decreases the size of the font.
- **Clear formatting:** removes all the existing formatting from the selected text.
- **Bold:** makes the text bold.
- **Italics:** italicises the text
- **Underline:** underlines the text.
- **Strikethrough:** enables you to draw a line through a word as if it has been crossed out.
- **Subscript:** set a letter below the text baseline.
- **Superscript:** set a letter above the line of text.
- **Change Case:** this button has a number of options to change the case of the text. This is very useful if you have accidentally typed in UPPER CASE and you need to change it all to lower case without retyping all the text.
- **Text Highlight Colour:** you use this option to highlight text in a variety of colours. This is just like using a highlighter pen on a printed page.
- **Font Colour:** here you can choose from a variety of colour options.

Text alignment

Change the way text aligns on the page using the alignment buttons in the Paragraph group on the Home tab.

- **Left align:** the text starts against the left margin, but is uneven on the right.
- **Right align:** the text is aligned against the right margin.
- **Centre:** the text is arranged evenly in the centre of each line with uneven margins on the left and right, depending on the length of each line of text.
- **Justify:** the spacing between the text is changed automatically so that the margins are straight on both the left and the right.

Preview a document.

- Print layout shows the document exactly as it will look when printed.
- Full screen reading shows the document in full screen view in an uncluttered screen.
- Web Layout shows how the document will look as a web page.
- Outline shows the structure of the document.
- Draft is a view in which you can quickly edit text.

PAGE LAYOUT

- **Portrait** – the paper is used so that the narrowest side is at the top, like in this book.
- **Landscape** – the paper is used so that the widest side is at the top.
- **Collated** – the pages of a document have been put together in the correct order per copy. **Insert Clip Art** Clip Art is a collection of drawings and multimedia that is installed with your office suite.

MICROSOFT EXCEL.

What is a spreadsheet?

- A spreadsheet is an electronic file made up of rows and columns.
- It helps us to sort and arrange data, and do calculations with the data, for example add all the numbers in a column.
- **Rows:** used to enter one record
- **Columns:** used to group types of information
- **Cell:** the place where a column and a row meet.

COMPONENTS OF A SPREADSHEET.

- **Workbook –** the name for a spreadsheet file that contains a number of worksheets.
- **Worksheet –** the page on which you enter data in a workbook. Or pages in a workbook.
- These pages are called **worksheets**
- A spreadsheet file is called **a workbook.**
- **A workbook consists of worksheets.**
- When you open a spreadsheet application, such as Excel, you will see a blank file known as a **WORKBOOK.**
- When you open a new workbook in Microsoft Excel there are usually three worksheets available. You can add more or delete the ones you do not need.
- **Address:** the column letter and row number of the active cell
- **Active cell:** the cell that is selected.
- **The Name box:** is an area that shows you the **address** of the **active cell** you are currently working in. This cell is known as the **active cell.**
- **A range:** a block of cells
- **The Formula bar:** the area when you can enter formulas to do your calculations.
- **Sort:** arrange items in a particular order.
- **To Merge cells:** means turning the selected cells into one cell.

- **Merge and Centre:** joins the cells into a single larger cell and centres the contents horizontally.
- **Merger Across:** joins all the cells in each row of the selection into a single cell.
- **Merge Cells:** simply merges the selected cells into a single larger cell.
- **Functions:** predefined formulas built into Excel.

COMMON FUNCTIONS

SUM: adds all the values in a range of cells.

PRODUCT: multiplies all the values in a range of cells

AVERAGE: calculates the average value in a range of cells

QUOTIENT: Divides one value by another value

COUNT: Counts the number of cells in a range of cells

MIN: Gives the lowest value in a range of cells

MAX: Gives the highest value in a range of cells.

SCALING: to expand or shrink a worksheet before its printed.

POWERPOINT

What is a presentation program?

- **A presentation program** is a multimedia software application that lets you combine text, graphics, animation (movement), tables, sound and video into a professional presentation to communicate a message, idea or any other information to a group.
- **A PowerPoint presentation** is a collection of pages or screens that are designed in a special way.
- **Presentation:** an activity in which a speaker explains an idea or piece of work to an audience.
- **Slide shows:** presentations in which you display a series of projected images, or slides.
- The presentations can be viewed as slides called a **slide show.**
- This is displayed on a large monitor or on a projected screen.
- This is a powerful presentation application that enables you to create professional – looking presentations quickly.
- **Slide:** a single page in a presentation. Or Each separate screen is called a **Slide.**
- Each page in a presentation is called **a slide**
- **Title slide:** the first page in a presentation.
- **Placeholder:** a container for text and graphic content on a slide.

PLANNING A PRESENTATION

Consider the following aesthetic, practical and ethical guidelines when designing a presentation.

- Identify the critical information for your presentation.

- The tone used to get the message across should differ according to the target audience.
- In a large room, you may need a projector and a microphone. Text legibility and background colours are also important.
- The presentation must be appropriate to the age group.
- The younger the audience, the shorter their attention span will be.
- The audience will lose interest if there are too many explanations of familiar subject content. They will also get bored and lose interest if there are too few explanations for those members who lack knowledge of the subject.
- Consider the value of using pictures rather than too much text, and of using charts rather than tables for figures.
- Do not use too many colours or effects as this distract the audience from what the slide is trying to say. Use contrasting background and text colours for easy readability. However, avoid colour combinations that will make it difficult for people with colour blindness to view the presentation.
- Plan the length of the presentation in line with the time available.
- Make sure that the presentation makes sense and that slides run in logical sequence.
- Consider the cultural background of the audience and adjust the content accordingly.

When creating the presentation, the following are critical:

- Consistent slide design throughout the presentation makes the message much cleaner.
- Limit the number of fonts used in the presentation to three. Keep the fonts consistent throughout the presentation.
- Use no more than six bullets per page. Bullets should be short ideas, not complete sentences.
- Do not type everything in capital letters. This is called SCREAMING and distracts the audience.

START POWERPOINT

- Click on the Start button
- Select All Programs.
- Scroll down the list of folders and open the Microsoft Office folder.
- Click the PowerPoint icon.
- When you start PowerPoint, you automatically get to choose a blank presentation. It will be called ***Presentation1***
- **A photo album** – is a presentation of personal or business photographs.
- **Animation –** using movement effects to focus on important points, to control the flow of information, and to increase viewer interest in a presentation.
- **Motion path –** in animation, the direction in which text or an object will make an entrance or exit.
- **Template:** a presentation that has already been designed and formatted.
- **Themes:** built in sets of colours, fonts and special effects.

DESKTOP PUBLISHING

- It involves using specially – designed software on a desktop computer to produce documents for publication.
- These include books, booklets, magazines, newspapers, newsletters, flyers, brochures, posters, cards and many more.
- This application is used to **typeset** the document.
- This means preparing a document to be printed by creating the page layouts.
- There are several well – known desktop publishing applications, often called **DTP applications for short.**

HOW DOES DTP DIFFER FROM WORD PROCESSING.

- ❖ Text and images are placed in frames that allow for blocks of content to be resized and rearranged easily. Moving text, and especially images, in a word processing document can have annoying side – effects with objects moving unexpectedly.

- Text can be set to flow automatically from one frame to another, if it needs to occupy more than one frame or must flow over onto another page.
- Typically, text is placed in columns on the page of a publication such as a newsletter. In DTP these columns are handled through frames, whereas a word processing document has actual columns on the page.
- **Header** – standard content found at the top or head of every page in a document.
- **Footer** – standard content found at the bottom or foot of every page in a document.
- **Publisher** does not immediately open onto a new blank document.
- You must first choose a template from which to work.
- To design your own layout, choose one of the blank templates.
- **Typography** – the style of text that has been typeset and prepared for printing.
- **Flyer** – an advertisement printed on a single page that is handed out to many people.
- **Frames**: containers for text and other objects on page.
- **Master page**: a page that has a layout you want to repeat on other pages.
- **Crop** – enables you to cut out part of the image.
- Black corners and lines appear on the edge of the picture.
- Drag them in and out to select how much of the image you want to keep.
- **Contrast** – increases or decreases the difference between the light and dark parts of an image.
- **Brightness** – determines how light or dark an image is.
- **Integration**: using data created in different applications in one document.

- **Gradient fill**: a gradual blending and fading of fill colours.